UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SCST1223
PROBABILITY AND STATISTICAL DATA ANALYSIS

PROJECT 2

LECTURER NAME
DR IZYAN IZZATI BINTI KAMSANI

SECTION 05

STUDENT NAME & MATRIC ID

RAJA NUR ALLEA DEWI MAHSURI BINTI RAJA MOHD YUSRI
A24CS0294

IRDINA SOFIA BINTI ROHAIDI
A24CS0253

NURZULAIKHA BINTI MOHD ISA
A24CS0291

MOHAMMED MOHSEN MOHAMMED ALSAKKAF
A23CS4026

Table of Contents

1.0 INTRODUCTION 3
2.0 DATASET 4
3.0 DATA ANALYSIS 5
3.1 HYPOTHESIS 2 SAMPLE TEST 5
3.1.1 Stress levels of attended counseling and Non-attended counseling 5
3.1.2 Hypothesis Statement 5
3.1.3 Test Statistics 6
3.1.4 Conclusion 6
3.2 CORRELATION TEST 7
3.2.1 Age and Stress Level 7
3.2.2 Hypothesis Statement 7
3.2.3 Analysis and Results 7
3.3 REGRESSION TEST 10
3.3.1 Peer Pressure and Mental Stress Level 10
3.3.2 Hypothesis Statement 10
3.3.3 Analysis and Result 10
3.4 CHI-SQUARE TEST 12
3.4.1 Hypothesis Statement 12
3.4.2 Analysis and Results 12
3.4.3 Decision and Conclusion 12
4.0 CONCLUSION 13
5.0 APPENDIX 14

1.0 INTRODUCTION

The problem of mental health in students has gained more and more attention, particularly in light of academic stress, social demands, and lifestyle problems. Stress is a common psychological reaction that may have a serious impact on academic achievements, emotional state, and the quality of life of students. It is therefore necessary to not only know the causes of stress among students but also the coping mechanisms that students use to deal with stress.

The dataset used in this study is called Student Mental Stress and Coping Mechanisms and was obtained from Kaggle, which offers detailed survey data of students. The data will include important variables that are associated with the level of mental stress, the coping mechanisms used, and other demographic variables, including age, gender, academic year, and marital status. Psychological and demographic data are combined in this dataset, which makes it especially appropriate to study the connections between stress and personal traits.

Some of the variables were chosen to be analyzed in more detail, such as the level of stress (as a numerical variable), the number of coping mechanisms applied, gender, age, and academic year. The selection of these variables was based on their applicability in the determination of the factors that affect the mental conditions and behavioral reactions of students to stress. Before statistical analysis, several data preprocessing procedures were performed, such as the elimination of missing entries and the conversion of categorical variables to an appropriate form of analysis.

This report aims to explore some of the important relationships, including the relationship between the number of coping strategies and stress levels, gender differences in stress experiences, and the predictive power of demographic variables on student stress. This study aims to provide useful information on the way students perceive and cope with stress through statistical analysis using t-tests, chi-square tests, correlation analysis, and linear regression, which will eventually lead to the establishment of improved support systems in educational institutions.

2.0 DATASET

Numerous variables were chosen for analysis. These include stress level, which is regarded as a numerical variable to quantify the level of stress that students experience, and coping mechanisms, which are the strategies that students employ to manage stress. The number of strategies that each student reported was used to further quantify coping mechanisms. Gender, academic year, and age were also chosen as significant variables, either for comparison between groups or as predictors in regression analysis. These variables were selected due to their direct correlation with the factors that are believed to affect student tension and their ability to facilitate the execution of a diverse array of necessary statistical tests.

Numerous data preprocessing procedures were implemented prior to the analysis. To preserve the reliability and accuracy of the results, entries that were either incomplete or absent were eliminated. Categorical variables, including gender and coping strategy types, were transformed into appropriate representations for statistical analysis. To facilitate correlation and regression analyses, coping mechanisms were quantified and organized. To guarantee lucidity throughout the analysis process, variable names and labels were also standardized.

Considering the chosen variables and the dataset's structure, multiple outputs are expected. A negative link is anticipated between the quantity of coping techniques and stress levels, indicating that students employing more coping mechanisms may encounter reduced stress. A notable disparity in stress levels may also exist between male and female students. Regression analysis may indicate that variables such as age can serve as predictors of stress levels. Finally, a chi-square test may reveal a correlation between gender and preferred coping mechanisms.

3.0 DATA ANALYSIS

3.1 HYPOTHESIS 2 SAMPLE TEST

3.1.1 Stress levels of attended counseling and Non-attended counseling

In this case, we explore whether there is a significant difference in the mean mental stress levels between students who attend counseling and those who do not. The 2-sample t-test is used since we are comparing the means of 2 independent groups with unequal and unknown variances.

The test statistic formula :

$$t_c = \frac{(\bar{x}_1 - \bar{x}_2) - \delta_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Where :

- $\bar{X}_1$, $\bar{X}_2$ are the sample means
- s_1, s_2 are the sample standard deviations
- n_1, n_2 are the sample sizes
- δ0=0 (we test for no difference)

Degrees of freedom are calculated using the **Welch-Satterthwaite equation**:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1 - 1}\left(\frac{s_1^2}{n_1}\right)^2 + \frac{1}{n_2 - 1}\left(\frac{s_2^2}{n_2}\right)^2}$$

3.1.2 Hypothesis Statement

H_0: μ1 = μ2

There is no significant difference in the mean stress level between students who attend counseling and those who do not.

H_1: μ1 ≠μ2

There is a significant difference in the mean stress level between students who attend counseling and those who do not.

3.1.3 Test Statistics

Using RStudio, the sample mean of mental stress levels for students who attended counseling was calculated as $\bar{X}_1 = 5.217$, while for students who did not attend counseling, the sample mean was $\bar{X}_2 = 5.504$. The sample standard deviations for these two groups were $s_1 = 2.850$ and $s_2 = 2.950$, respectively. The number of students who attended counseling was $n_1 = 373$, and the number who did not was $n_2 = 387$.

To test the hypothesis whether attending counselling influences students' stress levels, a **two-sample t-test** assuming unequal variances (Welch's t-test) was conducted in RStudio. The calculated **test statistic** was $T = -1.363$ and the corresponding **degrees of freedom** were $df = 758$. Additionally, the **p-value** obtained from the test was $p = 0.1733$, and the **95% confidence interval** for the difference in mean stress levels was between -0.700 and 0.1260.

Since the test statistic is not large enough in absolute value and the **p-value exceeds 0.05**, the result is not statistically significant. Also, the 95% confidence interval includes **zero**, indicating that the difference in mean stress levels between the two groups might not be real, but due to random variation.

3.1.4 Conclusion

Since the p-value $0.1733 > 0.05$, we **fail to reject** the null hypothesis $\mathbf{H_0}$. There is **insufficient evidence** to support the claim that attending counseling significantly changes the average stress level among students.

3.2 CORRELATION TEST

3.2.1 Age and Stress Level

To determine whether there is a statistically significant correlation between a student's age and their reported level of mental stress.

3.2.2 Hypothesis Statement

H_0: There is no correlation between students' age and stress level.

H_1: There is a significant negative correlation between students' age and stress level.

3.2.3 Analysis and Results

Using the Pearson correlation formula:

$$r = [\ \Sigma(x_i - \bar{x})(y_i - \bar{y})\ ] / \sqrt{[\ \Sigma(x_i - \bar{x})^2 \times \Sigma(y_i - \bar{y})^2\ ]}$$

x_i = values in the Age column

y_i = values in the Mental Stress Level column

Data Summary:

- Number of students(n) = 760
- Total Age($\sum x$) = 18179
- Total Stress level($\sum y$) = 4076
- Total Age² ($\sum x2$) = 445909
- Total Stress²($\sum y2$) = 28256
- Total Age × Stress($\sum xy$) = 97091

We use a simplified formula:

$$r = \frac{\{n \sum xy - \sum x \sum y\}}{\left\{\sqrt{\{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]\}}\right\}}$$

Substituting Values:

Numerator:

$$760(97091) - (18179)\ (4076) = 73829160 - 74126104 = -296944$$

Denominator:

$$\sqrt{\{[760(445909) - (18179)^2][760(28256) - (4076)^2]\}}$$

$$= \sqrt{\{(338888840 - 330481041)(21474560 - 16614076)\}}$$

$$= \sqrt{\{8407799 \cdot 4860484\}}$$

$$= \sqrt{\{40873242452116\}} \approx 6393986.2$$

Final Calculation:

$$\frac{-296944}{6393986.2} \approx -0.0464$$

To compute the p-value from the t-distribution table (df = n-2)

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

- r = -0.048
- n = 760

t = ≈ -1.33

- Then look up t = 1.33 with df = 758 → $p \approx 0.184$

OUTPUT

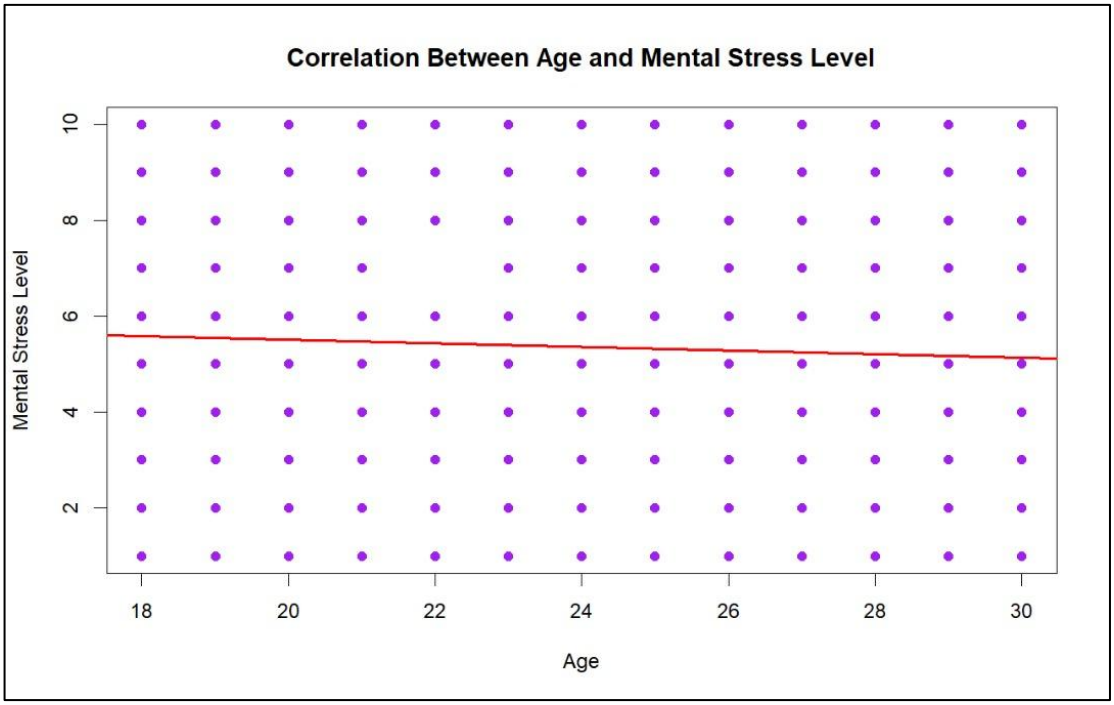


Figure 1: Correlation Between Age and Mental Stress Level

INTERPRETATION

- The correlation is approximately -0.048, indicating a very weak negative linear relationship between age and stress level.
- The p-value is 0.184, which is greater than the significance level ($\alpha = 0.05$).

The scatter plot illustrates the relationship between students' age and their reported mental stress levels. Each point on the graph represents an individual student, while the red line indicates the line of best fit. The slight downward slope of the trend line suggests a weak negative relationship between the two variables, indicating that as age increases, mental stress level tends to decrease slightly. However, the data points are widely scattered with no clear pattern, reinforcing that correlation is minimal. This observation aligns with the calculated Pearson correlation coefficient of -0.048, which indicates a very weak negative correlation. Overall, the plot supports the conclusion that there is no statistically significant linear relationship between age and mental stress level in this dataset. Since the p-value is greater than 0.05, we fail to reject the null hypothesis. Although a slight negative trend exists visually, which is that older students tend to report slightly less stress, the relationship is not strong enough to be statistically meaningful.

3.3 REGRESSION TEST

3.3.1 Peer Pressure and Mental Stress Level

To examine predictors of stress further, a simple linear regression was carried out to check whether peer pressure had any chance to represent variation in the mental stress levels significantly.

3.3.2 Hypothesis Statement

- **H_0:** There is no significant linear relationship between peer pressure and mental stress level among students.
- **H_1:** There is a significant linear relationship between peer pressure and mental stress level among students.

3.3.3 Analysis and Result

Data summary:

- Total Peer Pressure(x) = 2267
- Total Stress Level(y)= 4076
- Total (x^2) = 9231
- Total (y^2) = 28256
- Total (x*y) = 11980

b_1: $\frac{11980-\frac{2267\cdot4076}{760}}{8231-\frac{2267^2}{760}}$

: -0.1214

b_0: $5.3632-(-0.1214\cdot2.9829)$

: 5.7252

OUTPUT

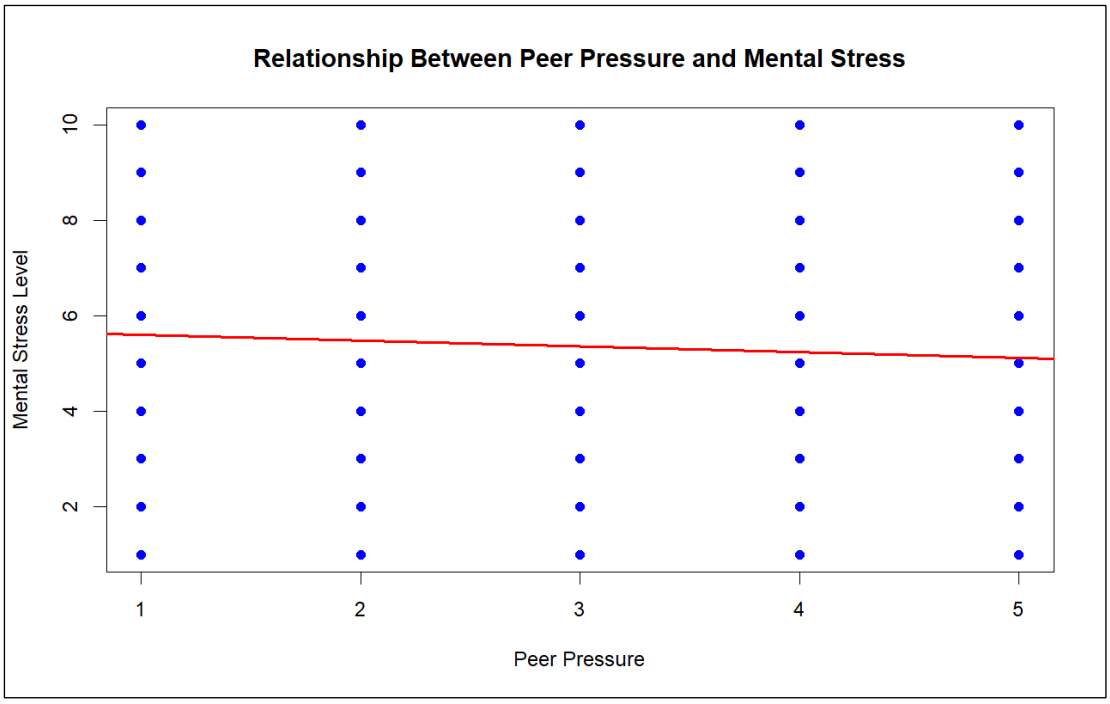


Figure 2: Relationship between Peer Pressure and Mental Stress

Interpretation

The scatter plot of the relationship between peer pressure and mental stress level is represented by a slight downward slope red regression line with a slight downward direction, which shows a very weak negative relationship between these two characteristics. The data points are distributed horizontally over all the levels of peer pressure, thus, there is no direct trend. Such a visual pattern, along with the statistical output, proves that the factor of peers' pressure cannot be called a serious and notable predictor of mental strain in this sample.

3.4 CHI-SQUARE TEST

To further explore the relationships within the dataset, a chi-squared test of independence was performed. The primary objective of this test was to determine if a statistically significant association exists between a student's gender and their preferred stress coping mechanism. This test assesses whether the choice of coping strategy is dependent on gender or if these two variables are independent.

3.4.1 Hypothesis Statement

- **H_0:** Coping strategy preferences are independent of gender (no association).
- **H_1:** Coping strategy preferences are associated with gender.

3.4.2 Analysis and Results

The dataset was filtered to include only "Male" and "Female" respondents to construct a clear contingency table. The chi-squared test was then applied to this table of observed frequencies.

The key results from the analysis in R are summarized below:

- **Test Statistic (χ^2):** The calculated Chi-Square value is **7.8159**
- **Degrees of Freedom (df):** The degrees of freedom for the test are **9**
- **P-value:** The resulting p-value is **0.5528**

The critical value for a Chi-Square test with 9 degrees of freedom at a significance level of
$\alpha = 0.05$ is **16.919**. An assumption check confirmed that all expected cell frequencies were greater than 5, validating the use of the Chi-Square test.

3.4.3 Decision and Conclusion

The p-value from our test is **0.5528**, which is significantly greater than the alpha level of 0.05. Furthermore, our calculated test statistic **($\chi^2 = 7.8159$)** is less than the critical value **(16.919)**.

Therefore, we **fail to reject the null hypothesis (H_0)**.

Based on this statistical decision, there **is not sufficient evidence** to conclude that an association exists between a student's gender and their choice of coping mechanism in this sample. The results suggest that the type of coping strategy a student employs is **independent** of their gender. Both male and female students appear to utilize the various coping mechanisms in proportionally similar ways.

4.0 CONCLUSION

This project looked closely at data from the **Student Mental Stress and Coping Mechanisms** study. The goal was to find important links between students' background (like age and gender), their stress levels, and how they deal with stress. We used three main types of tests: Correlation, Linear Regression, and Chi-Square to turn raw numbers into clear insights about student mental health. First, we used a **Correlation Test** to see if there was a link between students' age and their stress levels. The result showed a very small negative connection ($r = -0.048$), and the p-value was 0.184. Since this p-value is higher than 0.05, we cannot say there's a real relationship. In short, age does not have a meaningful effect on how stressed students feel, even though the graph showed a very slight trend. Next, we did a **Simple Linear Regression** to find out if **peer pressure** could predict how stressed students are. The result showed no strong link. The p-value was much higher than 0.05, so we can't say peer pressure is a useful way to predict stress. The data points were spread out widely, meaning that other things likely cause stress more than peer pressure. Lastly, we ran a **chi-squared test** to see if gender affects how students cope with stress. The test gave a p-value of 0.5528, which is way above 0.05. So, we found no link between gender and coping strategies. This means male and female students tend to use similar ways to deal with stress. Finally, this study shows that some common beliefs about student stress are not always true when tested with real data. Age and peer pressure don't have a big effect on stress levels, and gender doesn't change how students cope. These results show that schools should offer mental health help to all students, not just based on age, gender or peer pressure. This project was a good example of how we can use statistics to better understand real-life problems.

5.0 APPENDIX

YouTube Link: https://youtu.be/pfzZqtt-Blo

Dataset Link: https://www.kaggle.com/datasets/salahuddinahmedshuvo/student-mental-stress-and-coping-mechanisms